

Chat Application – Socket Programming

Test Plan

Team 5

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Overview

This document contains comprehensive test cases for the Chat Application using WebSocket technology. A total of 64 test cases have been designed covering Unit Testing (UT), Integration Testing (IT), and System Testing (ST) across 7 major functional modules.

MODULE 1: USER REGISTRATION

ID	Module	Description	Pre-Condition	Steps	Test Data	Expected Result
UT-REG-01	User Registration	Test valid registration with all required fields	Access to application	1. Navigate to registration page 2. Enter valid email, username, password 3. Click Register	Valid credentials	Account created, redirect to login
UT-REG-02	User Registration	Test registration with duplicate email	Email already registered	1. Navigate to registration 2. Enter existing email 3. Click Register	Duplicate email	Error: 'Email already registered'
UT-REG-03	User Registration	Test registration with duplicate username	Username already exists	1. Navigate to registration 2. Enter existing username 3. Click Register	Duplicate username	Error: 'Username already taken'
UT-REG-04	User Registration	Test registration with invalid email format	None	1. Enter invalid email 2. Click Register	notanemail	Error: 'Invalid email format'
UT-REG-05	User Registration	Test registration with weak password	None	1. Enter short password (<8 chars) 2. Click Register	Pass: 123	Error: 'Password too short'
IT-REG-01	User Registration	Test database insertion of new user	Database connected	1. Complete registration 2. Query database for user	New user data	User record created in database
ST-REG-01	User Registration	Test full registration to login flow	Application running	1. Register user 2. Login with new credentials	Complete flow	Successful login after registration
ST-REG-02	User Registration	Test SQL injection protection	Application running	1. Try SQL injection in email field 2. Submit form	SQL injection attempt	Input sanitized, no SQL execution

MODULE 2: USER AUTHENTICATION

ID	Module	Description	Pre-Condition	Steps	Test Data	Expected Result
UT-AUTH-01	User Authentication	Test login with valid credentials	Valid user exists	1. Enter username and password 2. Click Login	testuser01 / Test@1234	Successful login, dashboard shown
UT-AUTH-02	User Authentication	Test login with incorrect password	User exists	1. Enter valid username 2. Enter wrong password 3. Click Login	testuser01 / WrongPass	Error: 'Invalid credentials'
UT-AUTH-03	User Authentication	Test login with non-existent user	None	1. Enter non-existent username 2. Click Login	nonexistent / Pass123	Error: 'User not found'
UT-AUTH-04	User Authentication	Test empty credential fields	None	1. Leave username empty 2. Click Login	Empty fields	Validation error: 'Username required'
UT-AUTH-05	User Authentication	Test session token generation	User logged in	1. Verify token created 2. Check token stored	Login data	Valid JWT token generated
IT-AUTH-01	User Authentication	Test login event logging	Logging enabled	1. Perform login 2. Query login_logs table	User login	Login event logged with timestamp
ST-AUTH-01	User Authentication	Test brute force protection	Application running	1. Attempt 5+ wrong logins 2. Check account status	Multiple failed attempts	Account locked after 5 failures
ST-AUTH-02	User Authentication	Test session timeout	User logged in	1. Remain idle 30+ minutes 2. Attempt action	Idle session	Session expired, redirect to login

MODULE 3: ONE-TO-ONE CHAT INITIATION

ID	Module	Description	Pre-Condition	Steps	Test Data	Expected Result
UT-CHAT-01	One-to-One Chat	Test initiating chat with valid contact	Two users registered	1. Open contacts list 2. Click on contact name 3. Verify chat window opens	Contact: testuser02	Chat window opens with empty history
UT-CHAT-02	One-to-One Chat	Test chat with previous messages	Chat history exists	1. Open chat 2. Scroll up to view old messages	Existing chat	Previous messages displayed chronologically
UT-CHAT-03	One-to-One Chat	Test real-time message updates	Two users active	1. Send message from User A 2. Check User B's screen	Test message	Message appears instantly on recipient (<200ms)
UT-CHAT-04	One-to-One Chat	Test chat with blocked user	User blocked	1. Try to chat with blocked user	Blocked contact	Chat prevented, error shown
UT-CHAT-05	One-to-One Chat	Test chat close and reopen	Chat open	1. Close chat window 2. Reopen by clicking contact	Same contact	Message history intact
IT-CHAT-01	One-to-One Chat	Test WebSocket real-time connection	WebSocket server running	1. Send message 2. Monitor WebSocket events	Message data	Message transmitted via WebSocket
ST-CHAT-01	One-to-One Chat	Test end-to-end chat flow	Two users logged in	1. User A sends message 2. User B replies 3. Verify order	Message exchange	All messages delivered in order
ST-CHAT-02	One-to-One Chat	Test chat persistence after refresh	Active chat	1. Send message 2. Refresh browser 3. Check message history	Sent messages	Chat restored with history intact

MODULE 4: SEND TEXT MESSAGE

ID	Module	Description	Pre-Condition	Steps	Test Data	Expected Result
UT-MSG-01	Send Text Message	Test sending simple text message	Chat open	1. Type message 2. Click Send	Hello World	Message sent and displayed
UT-MSG-02	Send Text Message	Test sending empty message	Chat open	1. Leave field empty 2. Click Send	Empty string	Send disabled or error shown
UT-MSG-03	Send Text Message	Test message with special characters	Chat open	1. Type special characters 2. Click Send	@#\$\$%^&*()	Special chars preserved
UT-MSG-04	Send Text Message	Test sending long message	Chat open	1. Type 2000 char message 2. Click Send	Long text	Message sent with word wrap
UT-MSG-05	Send Text Message	Test character limit enforcement	Chat open	1. Type beyond limit 2. Try to add more	Over limit text	Input limited or warning shown
IT-MSG-01	Send Text Message	Test message storage in database	Message sent	1. Send message 2. Query messages table	Message content	Record created with timestamp
ST-MSG-01	Send Text Message	Test rapid message sending	Chat open	1. Send 5 messages quickly 2. Verify all delivered	5 messages	All messages in correct order
ST-MSG-02	Send Text Message	Test message with network issue	Chat open	1. Send message 2. Simulate disconnect 3. Check retry	Message during outage	Message queued or retried

MODULE 5: MESSAGE DELIVERY STATUS

ID	Module	Description	Pre-Condition	Steps	Test Data	Expected Result
UT-DELIV-01	Message Delivery	Test sending status indicator	Message transmitting	1. Send message 2. Observe status icon	Message being sent	Single checkmark shown
UT-DELIV-02	Message Delivery	Test sent status indicator	Message at server	1. Send message 2. Wait for acknowledgment	Server acknowledgment	Single checkmark persists
UT-DELIV-03	Message Delivery	Test delivered status indicator	Recipient online	1. Send message 2. Observe status change	Online recipient	Double checkmark displayed
UT-DELIV-04	Message Delivery	Test delivery to offline recipient	Recipient offline	1. Send message 2. Wait for recipient online	Offline user	Status changes to delivered when online
UT-DELIV-05	Message Delivery	Test failed delivery status	Delivery fails	1. Send message 2. Simulate failure	Failed delivery	Error indicator with retry option
IT-DELIV-01	Message Delivery	Test server acknowledgment	Message sent	1. Send message 2. Capture server ACK	Server response	ACK received with message ID
ST-DELIV-01	Message Delivery	Test complete delivery flow	Two users active	1. Send message 2. Monitor status progression	Complete flow	Status: sending → sent → delivered
ST-DELIV-02	Message Delivery	Test batch message delivery	Multiple messages	1. Send 3 messages 2. Check individual statuses	3 consecutive messages	Each tracked independently

MODULE 6: READ RECEIPTS

ID	Module	Description	Pre-Condition	Steps	Test Data	Expected Result
UT-READ-01	Read Receipts	Test read receipt on message view	Message delivered	1. Recipient opens chat 2. Message becomes visible 3. Check sender's view	Displayed message	Double checkmark turns blue
UT-READ-02	Read Receipts	Test no receipt if not viewed	Message not opened	1. Send message 2. Don't open chat 3. Check status	Unread message	Checkmark remains double blue
UT-READ-03	Read Receipts	Test multiple message reads	Multiple messages	1. Send 3 messages 2. Recipient reads all 3. Check indicators	3 messages	All show read status
UT-READ-04	Read Receipts	Test receipt from offline then online	Offline recipient	1. Send message 2. Recipient comes online 3. Reads message	Offline then online	Receipt sent after reading
UT-READ-05	Read Receipts	Test disabling read receipts	Privacy enabled	1. Disable read receipts 2. Send and read message	Disabled receipts	Read status not visible to sender
IT-READ-01	Read Receipts	Test WebSocket read event	Message read	1. Monitor WebSocket 2. User reads message	Read event	WebSocket emits read receipt
ST-READ-01	Read Receipts	Test end-to-end receipt flow	Two users active	1. Send message 2. Recipient reads 3. Verify receipt	Complete flow	Receipt delivered in real-time
ST-READ-02	Read Receipts	Test receipts with network delay	Network delay	1. Send and read with delay 2. Verify eventual delivery	Delayed network	Receipt delivered despite delay

MODULE 7: GROUP CHAT CREATION

ID	Module	Description	Pre-Condition	Steps	Test Data	Expected Result
UT-GROUP-01	Group Chat	Test creating group with valid members	Multiple contacts	1. Click Create Group 2. Enter name, select members 3. Click Create	3 members selected	Group created, window opens
UT-GROUP-02	Group Chat	Test group creation without name	None	1. Skip name field 2. Select members 3. Click Create	Empty name	Error: 'Group name required'
UT-GROUP-03	Group Chat	Test group with single member	Contacts available	1. Create group 2. Select only 1 member	1 member	Error: 'Minimum 2 members'
UT-GROUP-04	Group Chat	Test group with many members	Many contacts	1. Create group 2. Add 100 members	100 members	Group created successfully
UT-GROUP-05	Group Chat	Test duplicate group name	Group exists	1. Create group with same name	Duplicate name	Allowed with suffix or error
IT-GROUP-01	Group Chat	Test database group creation	Database active	1. Create group via UI 2. Query database	Group data	Records created in DB
ST-GROUP-01	Group Chat	Test end-to-end group messaging	3 users active	1. Create group 2. User A sends message 3. Verify received by B & C	Group workflow	All members receive message
ST-GROUP-02	Group Chat	Test group creation notifications	Group created	1. Create group 2. Check notifications for members	Notification delivery	All members notified

TEST EXECUTION SUMMARY

Total Test Cases: 64

Distribution by Type:

- Unit Testing (UT): Tests individual functions/modules
- Integration Testing (IT): Tests module interactions
- System Testing (ST): Tests complete workflows

Modules Tested (8 test cases each):

1. User Registration - 8 cases
2. User Authentication - 8 cases
3. One-to-One Chat - 8 cases
4. Send Text Message - 8 cases
5. Message Delivery Status - 8 cases
6. Read Receipts - 8 cases
7. Group Chat Creation - 8 cases

Instructions for Test Execution:

1. Execute each test case in the specified order
2. Document the actual results obtained
3. Compare against expected results
4. Mark as 'Pass' if match, 'Fail' if different
5. For failed tests, report issue and severity
6. Notify development team of findings

Additional Modules for Future Test Cases:

- Message Deletion and Editing
- File and Image Sharing
- Push Notifications
- Search Functionality
- User Blocking and Muting
- Voice and Video Calls